Topic 3: The Water Cycle & Water Insecurity

3.1 The Global Water Cycle

Hydrological cycle: The continuous movement of water between the atmosphere, land surface, and subsurface through evaporation, condensation, precipitation, runoff, infiltration, and groundwater flow. It is a *closed system* at the global scale (total water volume is fixed at ~1.386 billion km^3) but an *open system* at the drainage basin scale (water and energy cross the basin boundary).

Global Water Stores

Store	% of Total Water	Type	Residence Time
Oceans	96.5%	Saltwater	~3,600 years
Ice caps & glaciers	1.74%	Freshwater (frozen)	~10,000 years (Antarctica: up to 1 million)
Groundwater (total)	1.69%	Mixed (mostly saline deep groundwater; ~30% freshwater)	100–10,000 years
Permafrost	0.022%	Frozen ground ice	10,000+ years
Surface freshwater (lakes, rivers)	0.01%	Freshwater	Lakes: ~17 years; Rivers: ~2 weeks
Soil moisture	0.001%	Freshwater in soil pores	1–2 months
Atmosphere	0.001%	Water vapour, droplets, ice crystals	~10 days

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Biosphere	0.0001%	Water in living organisms	~1 week

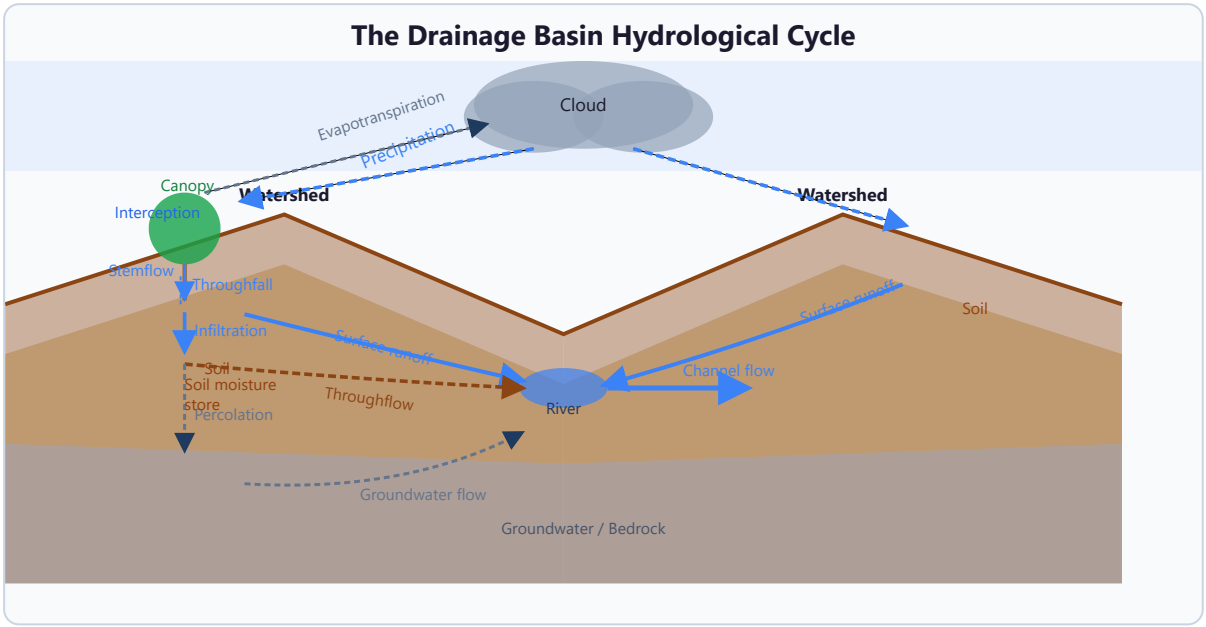
Only ~2.5% of Earth's water is freshwater, and of that, ~68.7% is locked in ice. Only ~0.3% of all freshwater is in accessible surface stores (rivers and lakes). This tiny fraction sustains all terrestrial life, agriculture, and industry.

Key Fluxes (Transfers)

- **Evaporation** — liquid water converts to vapour, driven by solar energy. ~505,000 km^3/yr evaporates globally (~86% from oceans). Rate increases with temperature, wind speed, surface area, and low humidity.
- **Transpiration** — water lost through plant stomata during photosynthesis. Tropical rainforests are the largest contributors: the Amazon returns ~8 trillion litres/day to the atmosphere. Transpiration links the water and carbon cycles.
- **Evapotranspiration** — combined evaporation and transpiration. Potential evapotranspiration (PET) is the theoretical maximum with unlimited water; actual evapotranspiration depends on water availability.
- **Condensation** — water vapour cools to dew point, forming droplets/ice crystals on condensation nuclei (dust, pollen, sea salt), creating clouds and fog.
- **Precipitation** — water falling from clouds as rain, snow, sleet, or hail. Three mechanisms: convectional (heated air rises), orographic/relief (air forced over mountains), frontal (warm air rises over cold air at a front). Global average: ~990 mm/yr over land.
- **Interception** — precipitation caught by vegetation canopy or built surfaces. In dense forest, 10–40% of rainfall may be intercepted and evaporated before reaching the ground.
- **Infiltration** — water soaking into soil from the surface. Rate depends on soil type (sandy: 25 mm/hr; clay: 1 mm/hr), antecedent moisture, vegetation, compaction, and frozen ground.
- **Percolation** — deeper downward movement through soil and rock into aquifers. Slow process through pores and fractures.
- **Throughflow** — lateral movement of water through the soil layer, usually following the slope gradient. Faster in sandy soils.
- **Groundwater flow** — very slow lateral movement through permeable rock (aquifers). May travel only metres per year. Sustains river baseflow during dry periods.
- **Surface runoff (overland flow)** — water flowing over the surface when rainfall intensity exceeds infiltration capacity (Hortonian overland flow) or when the soil is fully saturated (saturated overland

flow). The fastest route to river channels.

- **Channel flow** — water flowing in a defined river channel towards the sea. Discharge (Q) = cross-sectional area (A) × velocity (V), measured in cumecs (m^3/s).



3.2 The Drainage Basin as a Local Open System

Drainage basin: The area of land drained by a river and its tributaries, bounded by a **watershed** (ridge of high land). It is an open system with one main input (precipitation) and outputs (discharge to the sea, evapotranspiration). The **water balance equation** summarises the basin's hydrology:

$$P = Q + E \pm \Delta S$$

Where: P = precipitation, Q = discharge (runoff), E = evapotranspiration, ΔS = change in storage (soil moisture, groundwater, snow/ice). In wet seasons, P > E and storage fills (soil moisture recharge). In dry seasons, E > P and storage is drawn down (soil moisture deficit).

Factors Affecting the Drainage Basin Water Balance

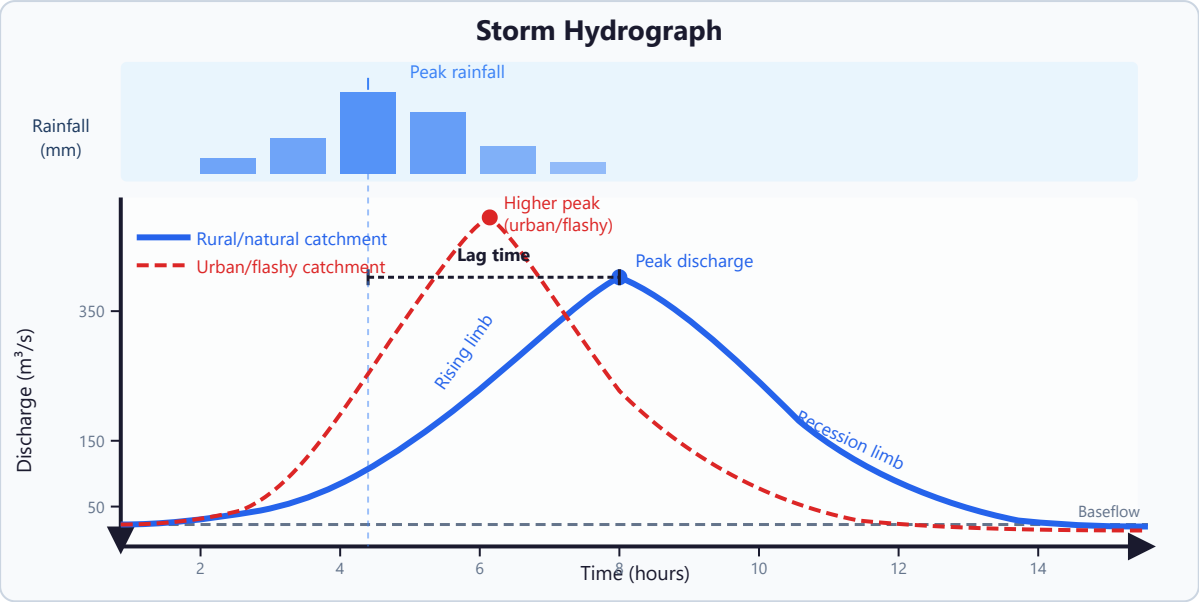
Factor	Effect on Water Balance
Climate	Rainfall amount and intensity, temperature (affects evapotranspiration rates), seasonality

Geology	Permeable rocks (chalk, limestone, sandstone) promote infiltration and groundwater storage; impermeable rocks (clay, granite) promote surface runoff
Soil type	Sandy soils: high infiltration, low runoff. Clay soils: low infiltration, high runoff, rapid saturation.
Vegetation	Increases interception (10–40%), transpiration, infiltration (root channels), and organic matter (improves soil water-holding capacity). Deforestation dramatically increases runoff.
Land use	Urbanisation: impermeable surfaces increase runoff 2–6x. Agriculture: may compact soil, increase runoff, or drain wetlands. Reservoirs: store water, regulate flow.
Relief/slope	Steep slopes: rapid surface runoff, short lag times. Gentle slopes: slower flow, more infiltration time.
Basin size and shape	Small/circular basins: short lag time, flashy hydrograph. Large/elongated: long lag time, attenuated peak.

The Storm Hydrograph

Storm hydrograph: A graph showing river discharge over time in response to a precipitation event. Key features: **peak discharge** (maximum flow), **lag time** (delay between peak rainfall and peak discharge), **rising limb** (increasing flow), **recession limb** (declining flow), and **baseflow** (sustained groundwater contribution).

A **flashy hydrograph** (steep rising limb, short lag time, high peak) indicates rapid transfer of water to the channel: impermeable surfaces, steep slopes, saturated/clay soils, urban areas, deforested catchments. A **flat hydrograph** (gentle rising limb, long lag time, low peak) indicates slow transfer: permeable geology, gentle slopes, dense vegetation, large basins.



Case Study: Boscastle Flash Flood, Cornwall (16 August 2004)
185 mm of rainfall fell in 5 hours over a small, steep catchment (23 km^2) underlain by impermeable slate bedrock. Soils were already saturated from previous wet weather. The River Valency's discharge reached ~180 m^3/s (normal: <1 m^3/s). **Impacts:** 58 properties flooded, 115 vehicles swept away, 4 buildings demolished, £30 million damage, but no deaths (rapid helicopter rescue of 100+ people). **Causes of flashiness:** small steep catchment, impermeable bedrock, saturated soils (antecedent conditions), intense convectional rainfall, narrow valley funnelling water. **Response:** £10 million flood defence scheme completed 2008 (channel widened and deepened, raised bridge, upstream culvert improvements, improved drainage).

3.3 Changes to the Water Cycle

Natural Factors

- **Seasonal variations** — monsoon climates experience dramatic seasonal shifts in precipitation (India receives ~80% of annual rainfall June–September).
- **El Nino/La Nina (ENSO)** — periodic changes in Pacific Ocean circulation causing global weather disruption. El Nino brings drought to Australia, SE Asia, and eastern Africa, and floods to western South America. La Nina brings opposite conditions. Recurrence: every 2–7 years.
- **Volcanic eruptions** — major eruptions inject aerosols into the stratosphere, reducing global temperatures and precipitation for 1–3 years (e.g. Pinatubo 1991).

- **Long-term climate change** — Milankovitch cycles (orbital variations) drive glacial-interglacial cycles, dramatically altering global water distribution between ice and ocean stores over 10,000–100,000-year timescales.

Human Factors

- **Climate change (anthropogenic)** — warming intensifies the water cycle: atmosphere holds ~7% more moisture per 1°C rise (Clausius-Clapeyron relation), increasing extreme rainfall intensity but also drought severity in arid regions. Accelerated glacier melt transfers water from cryospheric to oceanic stores.
- **Deforestation** — removes interception and transpiration, increasing surface runoff by 20–40%. Reduces atmospheric moisture recycling (the Amazon generates ~50% of its own rainfall through transpiration). Increases flood risk and soil erosion.
- **Urbanisation** — impermeable surfaces (40–90% of urban land) reduce infiltration and groundwater recharge while increasing surface runoff speed and volume. London's paved front gardens increased from 28% to 48% between 1998 and 2008.
- **Water abstraction** — excessive groundwater pumping depletes aquifers faster than natural recharge. Ogallala Aquifer (US Great Plains): water table declined 30+ m in parts of Texas and Kansas since 1950; may be exhausted within 50 years in some areas.
- **Dam construction** — alters natural flow regimes, reduces downstream sediment supply, increases evaporation from reservoir surfaces, and disrupts aquatic ecosystems. ~58,000 large dams worldwide.
- **Agricultural irrigation** — ~70% of global freshwater withdrawals. The Aral Sea shrank to ~10% of its 1960 size due to irrigation abstraction from feeder rivers (one of the world's worst environmental disasters).

3.4 Water Insecurity

Water insecurity: The inability to access sufficient, safe, and affordable water for health, livelihoods, and well-being. The UN defines **water scarcity** as <1,000 m^3/person/year and **water stress** as <1,700 m^3/person/year. By 2025, 1.8 billion people will live in regions with absolute water scarcity; by 2050, over half the world's population could face water stress.

Causes of Water Insecurity

- **Physical scarcity** — insufficient water resources due to aridity, unreliable rainfall, or depleted sources. Affects the Middle East, North Africa, Central Asia, parts of India, Australia, and western USA.
- **Economic scarcity** — water exists but communities lack infrastructure, investment, or governance to access it. Affects much of Sub-Saharan Africa (water sources exist but treatment plants, pipes, and wells are inadequate). 2 billion people lack safely managed drinking water (2023).
- **Population growth** — global water demand has increased 600% over the past century. More people = more domestic, agricultural, and industrial demand.
- **Pollution** — industrial discharge, agricultural runoff (fertilisers cause eutrophication; pesticides contaminate drinking water), untreated sewage (80% of global wastewater is discharged untreated), and plastic waste reduce usable freshwater supply.
- **Climate change** — shifting precipitation patterns, glacier retreat (1.9 billion people depend on glacial meltwater), increased evaporation, more frequent and severe droughts.
- **Governance failures** — corruption, lack of regulation, inequitable pricing, poor maintenance of infrastructure, failure to invest in water treatment.
- **Transboundary conflict** — 263 rivers cross international borders; 60% lack cooperative management frameworks. Water disputes can escalate geopolitical tensions.

Consequences of Water Insecurity

- **Health** — waterborne diseases (cholera, typhoid, dysentery) kill ~485,000 people/yr. Contaminated water causes 1.2 million deaths annually (WHO). Women and girls in LICs spend an estimated 200 million hours/day collecting water, reducing time for education and economic activity.
- **Food production** — agriculture consumes 70% of freshwater. Water scarcity directly threatens food security: irrigated land produces 40% of global food from just 20% of cropland.
- **Economic development** — industry requires water (manufacturing, cooling, processing). Water insecurity constrains industrial development and GDP growth.
- **Conflict** — competition for scarce water resources can trigger or exacerbate conflict. The Syrian civil war was preceded by a severe drought (2006–2010) that displaced 1.5 million rural people to cities, contributing to social unrest.
- **Environmental degradation** — over-abstraction depletes wetlands, lowers water tables, causes land subsidence (Mexico City sinks 50 cm/yr due to aquifer depletion), and destroys aquatic ecosystems.
- **Migration** — water scarcity drives displacement. The World Bank estimates 216 million people could be forced to migrate within their countries by 2050 due to water stress and climate change.

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Strategies to Address Water Insecurity

Strategy	Description	Advantages	Disadvantages
Dams and reservoirs	Store water for supply, irrigation, hydropower, flood control	Reliable supply, renewable energy, flood management	Displacement (Three Gorges: 1.3M people), ecosystem damage, sedimentation, evaporation losses, high cost ($billions), earthquake risk from reservoir loading
Desalination	Removing salt from seawater via reverse osmosis or thermal distillation	Unlimited raw material, reliable regardless of climate. Saudi Arabia: 50% of water from desalination. Israel: 80% of domestic water.	Energy-intensive (3–10 kWh/m^3), expensive ($0.50–3.00/$m^3$), brine disposal harms marine life, carbon footprint unless solar-powered
Water transfer schemes	Moving water from surplus to deficit regions via pipelines, canals, or aqueducts	Addresses regional imbalance. China's South-North Transfer: 44.8 billion m^3/yr capacity, $79 billion.	Extremely expensive, political conflict between donor and recipient regions, environmental damage at source, evaporation losses
Rainwater harvesting	Collecting and storing rainwater from roofs or ground surfaces	Low-cost, decentralised, community-managed. Effective in monsoon climates (India, Bangladesh).	Seasonal, limited volume, quality concerns (contamination from roofs), storage costs
Water recycling/reuse	Treating wastewater for agricultural, industrial, or potable reuse	Reduces freshwater demand, reduces pollution. Singapore's NEWater: 40% of water supply from recycled wastewater. Windhoek (Namibia): direct potable reuse since 1968.	Public perception resistance, treatment energy costs, pharmaceutical contaminant concerns

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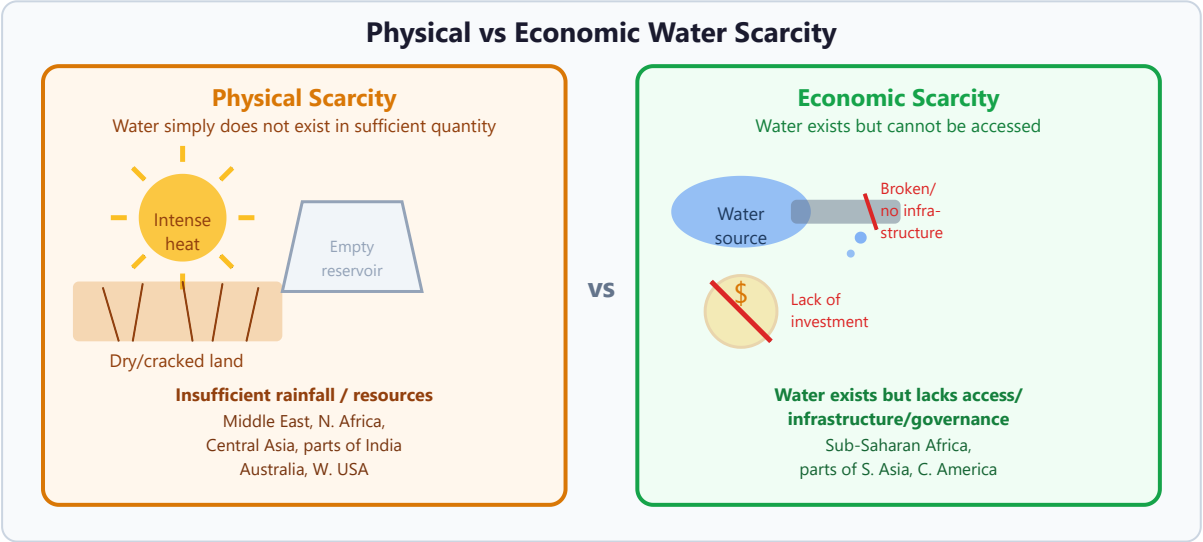
Demand management	Metering, pricing, leak repair, efficient irrigation, behaviour change campaigns	Cost-effective, reduces waste. Water metering reduces use 10–15%. Drip irrigation uses 30–50% less water than flood irrigation.	Requires behaviour change (difficult), pricing may harm the poor, infrastructure investment needed
Appropriate technology for LICs	Hand pumps (India Mark II), biosand filters, solar disinfection (SODIS), fog nets, protected springs	Low-cost, locally maintainable, empowers communities, culturally appropriate	Limited scale, maintenance challenges, may not meet growing urban demand



Case Study: The Aral Sea — Human-Caused Water Disaster

The Aral Sea (Central Asia, Kazakhstan/Uzbekistan) was once the world's 4th largest lake (68,000 km^2). Soviet-era irrigation projects from the 1960s diverted the Amu Darya and Syr Darya rivers to irrigate cotton fields in the desert. **Consequences:** the Aral Sea shrank to ~10% of its original size by 2014. The exposed seabed became a salt desert, with toxic dust storms carrying pesticides and salt across hundreds of kilometres, causing respiratory disease and cancer in surrounding populations. The local fishing industry collapsed (60,000 jobs lost). Climate became more extreme (hotter summers, colder winters) without the moderating lake effect. Biodiversity devastated. **Partial recovery:** Kazakhstan built the Kok-Aral Dam (2005, $86 million, funded by World Bank) to separate the North Aral Sea, allowing it to refill. The North Aral has recovered significantly (surface area

increased 40%, salinity decreased, fish returned). However, the South Aral remains largely dry and recovery is considered impossible without abandoning cotton irrigation. The Aral Sea is a stark warning about the consequences of unsustainable water abstraction.

Case Study: Water Conflict — The Nile Basin

The Nile (6,650 km, 11 countries) is the world's most geopolitically sensitive transboundary river. Egypt depends on the Nile for 95% of its freshwater and claims "historic rights" (55.5 billion m^3/yr under 1959 treaty). Ethiopia's Grand Ethiopian Renaissance Dam (GERD) on the Blue Nile (which provides 85% of Nile flow) has a 74 billion m^3 reservoir. **Ethiopia's argument:** the dam will generate 6,000 MW of electricity, lifting millions from energy poverty (65% lack electricity). **Egypt's fears:** reduced downstream flow could devastate 3.5 million hectares of irrigated agriculture and threaten water supply for 107 million people. Egypt is already below the UN water scarcity threshold (<500 m^3/person/year). **Sudan:** caught between benefits (cheaper electricity, flood control) and risks (dam failure, reduced silt for agriculture). Negotiations under the African Union have repeatedly stalled. The dispute illustrates how transboundary water creates geopolitical tension, particularly when combined with rapid population growth (both countries' populations are growing ~2%/yr), climate change (reduced Nile flows projected), and competing development priorities.

Case Study: UK Water Management

The UK faces a paradoxical water challenge: overall, the UK receives abundant rainfall (~1,126 mm/yr national average), but supply and demand are spatially mismatched. The wettest areas (NW Scotland: >3,000 mm/yr) have the lowest population density, while the driest areas (SE England: <600 mm/yr, comparable to Marrakech) have the highest demand (London: 9 million people). The Environment Agency warned in 2019 that England could face water supply shortfalls by 2050 without action. **Challenges:** population growth in SE England, climate change (drier summers projected), ageing infrastructure (UK water companies lose ~3 billion litres/day to leaks — ~20% of treated supply), over-abstraction of chalk aquifers (some rivers in SE England are drying up), pollution from agriculture (nitrate contamination). **Strategies:** Thames Tideway Tunnel (£4.3 billion "super sewer" to reduce sewage overflow into the Thames), proposed Abingdon Reservoir (controversial £2.6 billion reservoir in Oxfordshire), water efficiency campaigns, compulsory metering (only ~55% of UK households have meters), and proposed national water transfer grid connecting wet and dry regions. The 2022 drought and hosepipe bans highlighted the urgency of long-term planning.

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Exam Tip: Edexcel expects you to understand the water cycle as a SYSTEM with interconnected stores and flows. When discussing human impacts, always explain the systemic effects: deforestation reduces interception and transpiration (reducing atmospheric moisture recycling), increases surface runoff (altering the hydrograph), and reduces infiltration (depleting groundwater stores). Think in terms of inputs, outputs, stores, transfers, and feedback.

Exam Tip: For water insecurity questions, always distinguish between physical and economic water scarcity and give named examples. Evaluate strategies by considering: effectiveness, cost, environmental impact, sustainability, equity (who benefits?), and scalability. The "best" solution depends on the context: desalination works for wealthy arid countries (Saudi Arabia, Israel) but is inappropriate for rural Sub-Saharan Africa where appropriate technology (hand pumps, biosand filters) is more effective.

Exam Tip: When writing about the water cycle and climate change, use specific data: "the atmosphere holds ~7% more moisture per 1°C of warming (Clausius-Clapeyron relation)," "Arctic glaciers have lost ~9,625 billion tonnes of ice since 1961," "global sea levels are rising at 3.6 mm/yr." Specific data demonstrates detailed knowledge and earns higher marks than vague statements about "more rainfall" or "melting ice."

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